

House Price Prediction – Project Summary

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, parking spaces, furnishing status, and other amenities. The dataset used for this project was obtained from Kaggle and analyzed using Python and machine learning libraries.

Data Preparation

The dataset was first explored to understand its structure and identify any data quality issues. Missing values and duplicate records were checked and handled appropriately. Categorical variables were converted into numerical form using one-hot encoding. To improve model performance and focus on relevant predictors, features with stronger correlations to house price were retained for model training.

Model Development

The dataset was divided into training and testing sets using an 80:20 split. Two regression models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

The models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	\$964,971.86	\$1,339,904.16	0.6448
Random Forest Regressor	\$1,007,505.10	\$1,372,854.17	0.6271

Key Findings

The analysis revealed that property area, number of bathrooms, and air conditioning availability were among the most influential factors affecting house prices. Larger properties with better amenities generally commanded higher market values.

The Linear Regression model achieved the best overall performance with an R^2 score of 0.6448, slightly outperforming the Random Forest Regressor, which achieved an R^2 score of 0.6271. This suggests that the relationship between the selected features and house prices is largely linear within this dataset.

Conclusion and Recommendation

The developed models were able to predict house prices with reasonable accuracy and provide valuable insights into the factors influencing property values. One interesting observation was that practical amenities such as air conditioning had a greater impact on house prices than some additional features like guest rooms or basements.

Based on these findings, real estate businesses and investors should prioritize properties with larger areas, multiple bathrooms, and modern amenities, as these characteristics consistently contribute to higher property valuations and stronger market demand.